CHAPTER 13

Cosmic Radiation and Radioelements in Nature

Contents

13.1. Primary Cosmic Radiation 374
13.2. Secondary Reactions in the Earth's Atmosphere 376
13.3. Radioelements in Nature 377
13.4. Cosmogenic Radionuclides 378
13.4.1. Survey 378
13.4.2. Tritium 379
13.4.3. 14-Carbon 380
13.5. Primordial Radionuclides 381
13.5.1. Very long-lived nuclides lighter than lead 381
13.5.2. Elements in the natural radioactive decay series 382
13.6. Radium and Radon in the Environment 387
13.7. Disequilibrium 388
13.8. Age Determination from Radioactive Decay 390
13.8.1. Dating by the ^{14}C method 390
13.8.2. Dating by K-Ar method 393
13.8.3. Dating by Rb—Sr method 394
13.8.4. Dating by the Re—Os method 395
13.8.5. Dating based on ^{238}U decay 396
13.9. Natural Radioactivity of the Oceans 398
13.10. Anthropogenic Radioactivity in Nature 398
13.10.1. Nuclear weapons 399
13.10.2. Nuclear power plant accidents 399
13.10.3. Releases from nuclear power plants 402
13.10.4. Other anthropogenic additions 402
13.11. Exercises 403
13.12. Literature 404

The early workers in nuclear science found that their measuring equipment had a constant background level of radiation which could be eliminated only partially even with the aid of thick shielding walls of iron and lead. It was assumed initially that this radiation had its origin in naturally radioactive elements present in the materials in the laboratory. However, in 1911 Hess carried measuring equipment into the atmosphere with the aid of balloons and learned that some background radiation increased with altitude. Obviously, at least a component of the laboratory background radiation had its

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origin in some extra-terrestrial source. Test equipment carried outside of the earth's atmosphere by rockets has given us data which provide a fairly accurate picture of the composition of the radiation that comes to the earth from space.

The investigation of cosmic radiation has had a profound influence on nuclear science. When Chadwick in 1932 discovered the neutron, the picture of matter seemed complete: all matter appeared to be composed of four fundamental particles: protons, neutrons, electrons, and photons. The existence of the electron neutrino was predicted by Pauli in 1927 but it was not proven until 1956. However, through studies of the cosmic radiation Anderson discovered the *positron* (the first *antiparticle*) in the same year. Five years later Anderson and Neddermeyer discovered another new particle with a mass about one-tenth of a proton or about 200 times heavier than the electron. This particle is the *muon*, designated by μ. Since that time a large number of subatomic particles have been discovered.

13.1. PRIMARY COSMIC RADIATION

A rather small fraction of the cosmic radiation consists of electromagnetic radiation and electrons. The former vary in energy from a small percentage of γ-rays to a considerable intensity of X-rays, to visible light and to radiation in the radiofrequency region. The types and intensities of this radiation have been of great importance to development of models of the formation and composition of the universe.

The major part of the cosmic radiation is nuclear particles with very high energy: approximately 70% protons, 20% α-particles, 0.7% lithium, beryllium, and boron ions, 1.7% carbon, nitrogen, and oxygen ions, the residual 0.6% ions of $Z > 10$. These ions are bare nuclei prior to interaction since their kinetic energies exceed the binding energies of all of the orbital electrons.

The cosmic particle radiation can be divided by energy into two major groups (Fig. 13.1). One group has energies mainly below 1 GeV and consists primarily of protons. This group originates mainly from the sun. Its intensity varies in relation to solar eruptions since at the time of such an eruption a large amount of solar material, primarily hydrogen, is ejected into space.

The second group has energies up to about 3×10^{11} GeV, although the intensity of the particles decreases with increasing energy, following the relation $N(E) \propto E^{-1.6}$, where $N(E)$ is the number of particles with energies in excess of E. Thus particles of 10^3 GeV have an intensity of about 10^{11} higher than particles of 10^{10} GeV. Within this high energy group the particles at the lower end of the energy spectrum are assumed to originate from sources within our galaxy (the Milky Way), while the particles of the higher energy end are assumed to come from sources outside of our galaxy. Different hypotheses, which for the most part are untested, suggest that the particles come from astronomical radio sources, exploding super novae, or colliding galaxies, etc. At least a